מבוא לעיצוב תצוגה
וחווית משתמש

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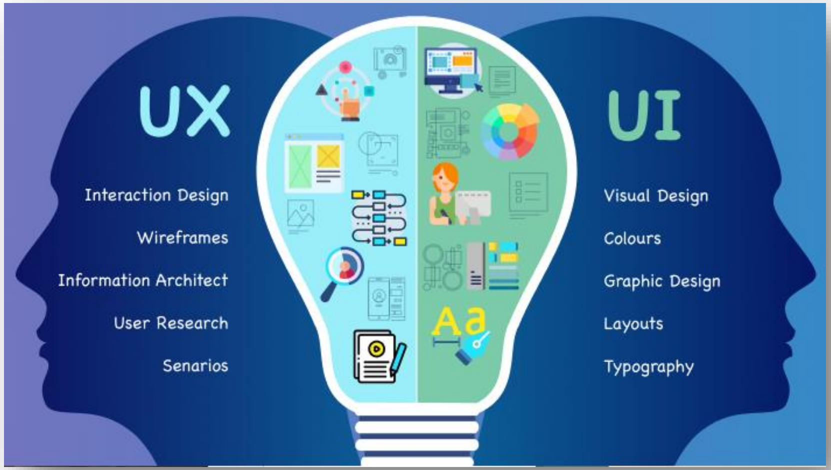
המרכז
האקדמי
לב

תחומים ותפקידים

מגוון מושגים

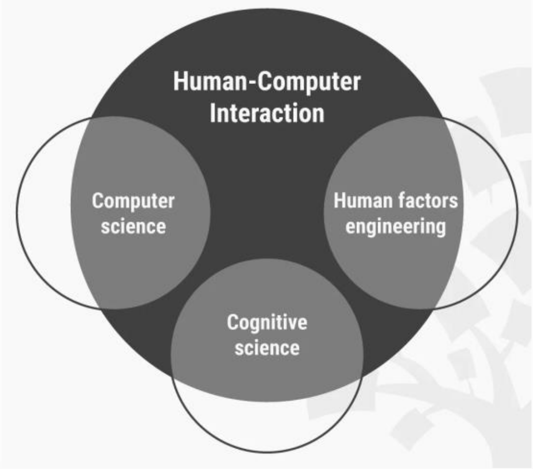
- Human Computer Interaction (HCI)
- Human Factors
- User Experience (UX)
- User-Centered Design
- Usability
- Interaction Design
- Information Architecture
- UI & Visual Design



HCI

Human-Computer Interaction refers to the study of
how humans and computers interact

- Generic Term that initially was concerned with computers, with time it expanded to cover almost all forms of information technology design.
- HCI professional may be a researcher, a designer, a psychologist, or anyone who might focus on human-computer interaction as part of their work or study.



https://www.interaction-design.org/literature/topics/human-computer-interaction

Human Factors

Multidisciplinary field incorporating contributions from psychology, engineering, industrial design, graphic design, statistics, operations research and anthropometry

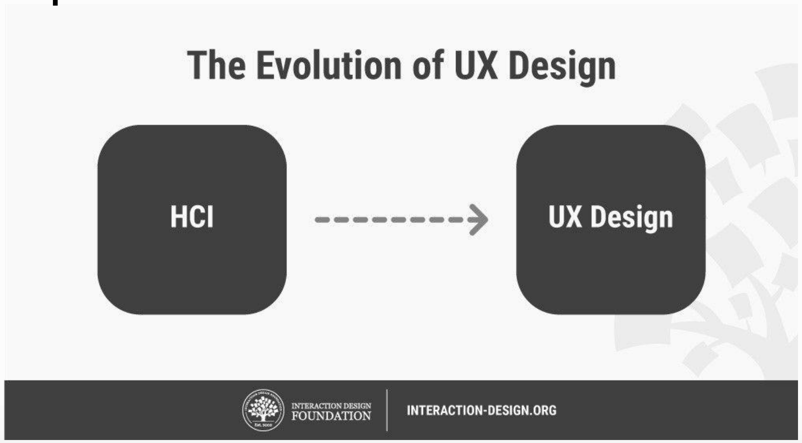
- Scientific understanding of the properties of human capability (*Human Factors Science*).
- Application of this understanding to the design, development & deployment of systems and services (*Human Factors Engineering*).
- The art of ensuring successful application of Human Factors Engineering to a program (sometimes referred to as *Human Factors Integration*) : also called ergonomics

User Experience (UX)

The overall experience a user has with a product

- UX is a field of work dedicated to helping a user make their way through a digital process or product with minimal effort and maximum value.

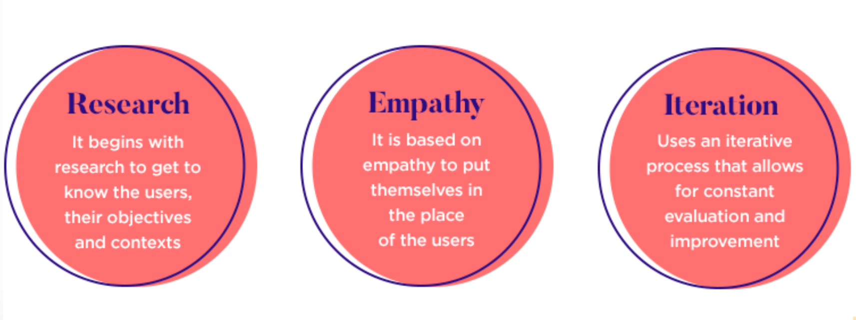
- This experience doesn't stop at the use of the product but starts with the advertising and presenting of the brand, through the purchase to the use and the day-to-day feeling the user carries with them about the product.



User Centered Design

A class of methodologies where design decisions are based on some tangible user model.

- User-centered design is the path to grate UX design
- Principels:
  - No point designing GUI without a clear *understanding* of intended user's profile – designers should set out to create a product that reflects the user needs and preferences.
  - Don't design for you – Design for your users
  - UCD is not linear. Create several versions of the same screen, test with real users, learn and perfect.



https://www.justinmind.com/blog/user-centered-design/

Usability

The ability of a specific type of user to be able to effectively carry out a task using a product

- Central Part of designing UX
- Usually measured through testing
- Number of test subjects that reflects the type of user that will use the application
- Test Measurements:
  - How many successfully complete a task.
  - On average, how quickly do they complete that task.
  - On average how many user errors are made while attempting to complete that task.

Two Additional Aspects of UX

- **Interaction Design:** The specific choices of user interactions we make to allow users to meet their goals in the software.
- **Information Architecture:** Structuring information so it best supports consumption by its target users.
- The two are considered **counterparts**
  - Interaction Designers focus on how people use computers to accomplish work
  - Information Architects focus on how people leverage information to support goals

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דגשים למימוש UX

Know the Users

- **Actor & Goal:** Often a job title or the common name for the type of user in a system
- **User Role:** Short name describing a user in pursuit of a goal. users change roles as their goals change
- **User Profile:** Summary information about the types of users who fill a role or perform as an actor begins a process of “profiling”
- **Persona:** Choosing specific characteristics of a person and compiling those into an archetypal description of that person creates a strong design target

Evaluate Usability

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- Usability is measured through testing Users placed in front of a prospective system are asked to perform specific tasks without instruction or guidance

- Typical usability tests measure:
  - Task completion frequency
  - Task completion time
  - Errors or mis-steps

- Professionals can give their subjective evaluation on usability, but you can't really be sure until you test or ship

Visual Design

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The design of the visual appearance of software or other commercial products.

- Visual Design must integrate with UX
- Visual design is concerned with the surface and overall feel of a design.
- Visual Design focuses more on esthetics
- Visual Designers are often NOT User Centered Designers

עקרונות מימוש UI

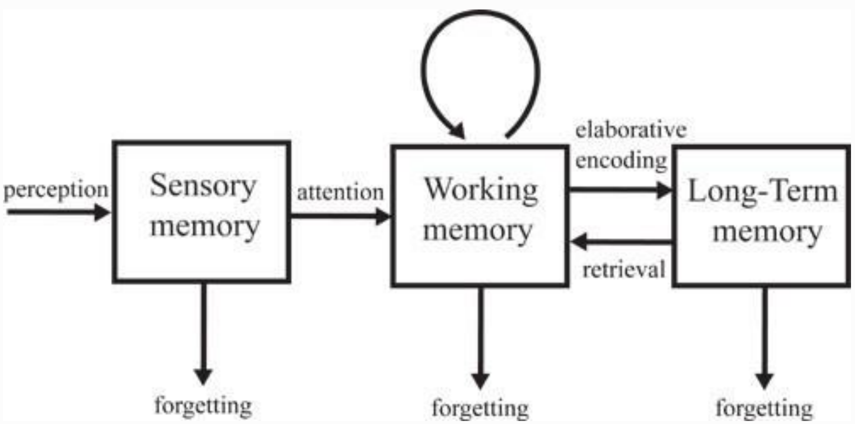
פסיכולוגיה קוגניטיבית

- תחום בפסיכולוגיה העוסק בהבנת תהליכים מנטליים
  - תהליכים מנטליים: חשיבה, קבלת החלטות, פתרון בעיות, שפה, קשב וזכרון
  - הגדרת יחידת ידע (קונספט, אב טיפוס, סכימה)
  - כיצד בני אדם לומדים - שיטות, מאפיינים ומגבלות
  - התייחסות דומה למחשב : בני אדם קולטים, מעבדים ומאחסנים אינפורמציה
  - גישה הפוכה לבהיוויוריזם

Cognition: “All processes by which the sensory input is transformed, reduced, elaborated, stored, recovered, and used.” *1967, Ulric Neisser*

דוגמא : Short-term memory

- הזכרון בו נשמרת אינפורמציה במהלך עיבוד מודע
  - זכרון זמני
  - חלקו מוקדש לאינפורמציה ויזואלית
  - מוגבל בנפחו
- משמעויות אפשריות:
  - אין להציף מידע במסך אחד : פיצול או יצירת היררכיה
  - יצירת תצוגה עקבית



Designing for Human Memory. How we can design interfaces that… | by Martin Jancik | UX Planet

Visual encoding : דוגמא

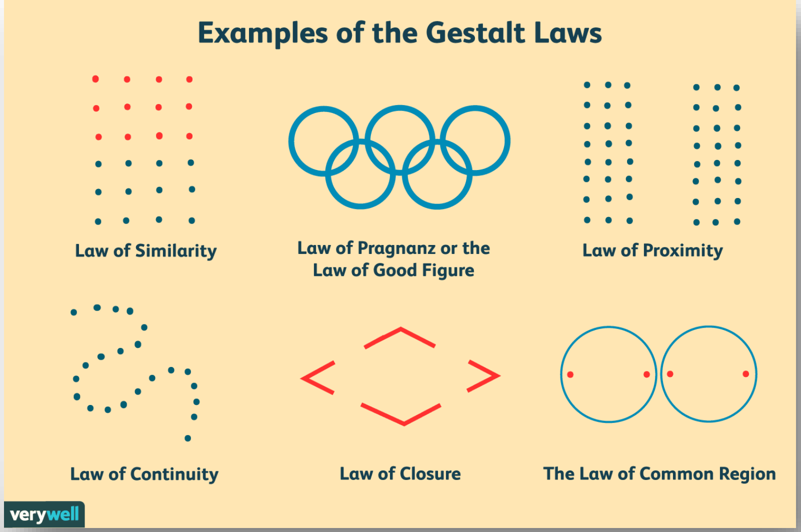
987349790275647902894728624092406037070570279072
8032080290073025012702370083740820787202727007083
247802602703793775709707377970667462097094702780
927979709723097230979592750927279798734972608027

98734979027**5**647902894728624092406037070**5**70279072
8032080290073025**0**12702370083740820787202727007083
24780260270379377**5**709707377970667462097094702780
927979709723097230979**5**927**5**0927279798734972608027

[Understanding Visual Encoding | Boost Labs]()

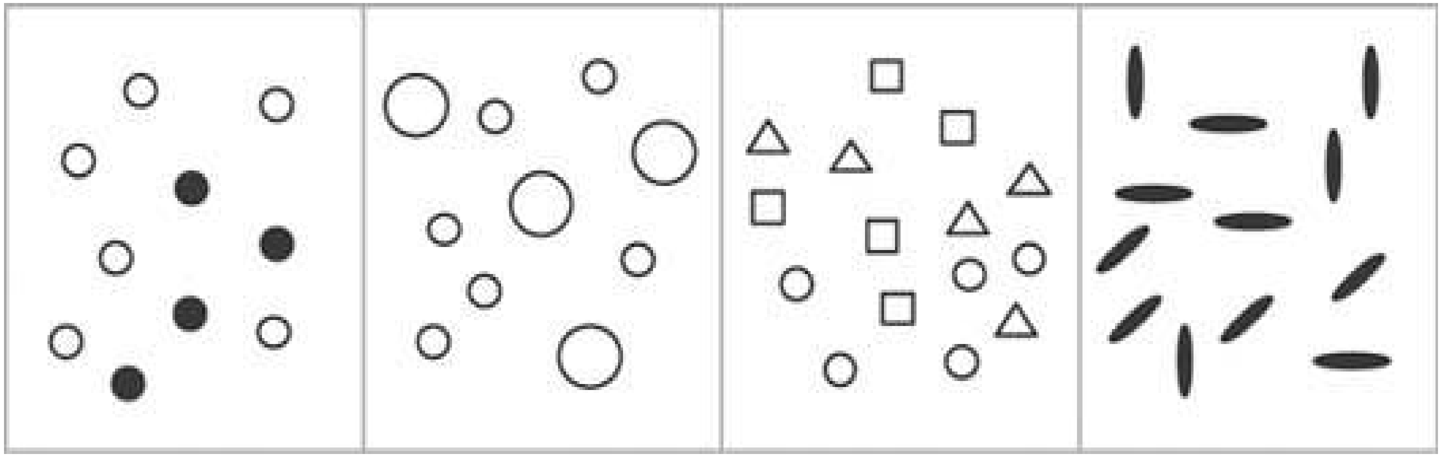
פסיכולוגית גשטלט

- גשטלט=תבנית
- האופן שבו אנו תופסים תבניות וצורה, הדרך שבה אנו מארגנים את שאנו רואים
- תובנות שימושיות לעיצוב תצוגה, בפרט ליצירת קשר בין פרטי מידע , הפרדה בין פרטי מידע או הבלטתם
- מציעה מספר עקרונות לעיצוב ויזואלי ובניהם:
  - Law of Proximity
  - Law of Closure
  - Law of Similarity
  - Law of Continuity
  - Law of “good figure”
  - Law of Common Region
  - Law of Connectedness



Law of Similarity

Similar things tend to appear grouped together

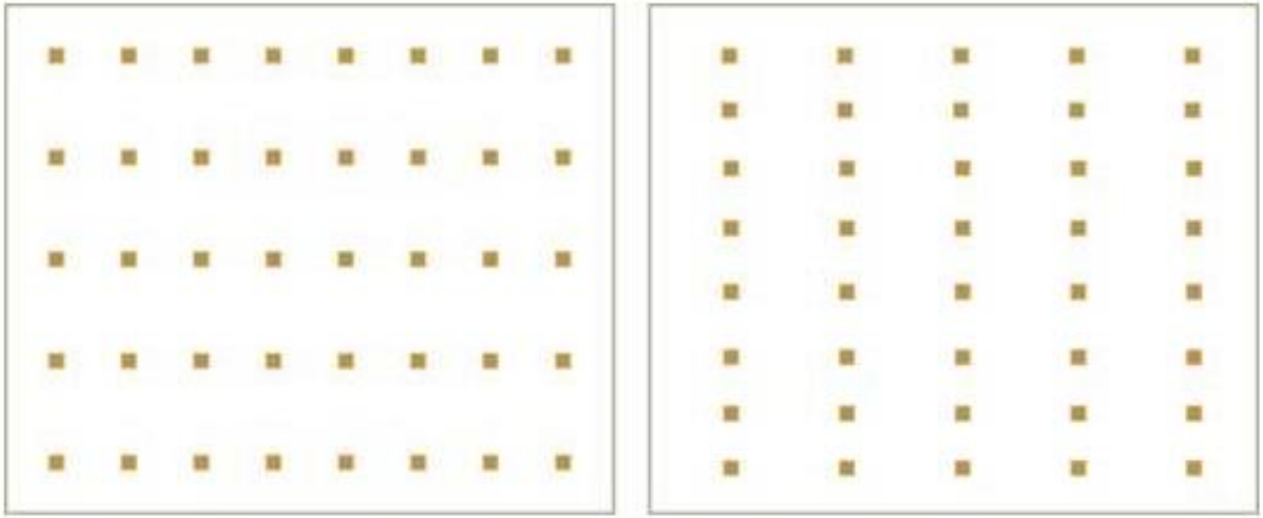


Law of "good figure"

When presented with a set of ambiguous or complex objects, they will be interpreted to appear as simple as possible

Law of Proximity

Things that are close together seem more related than things that are spaced farther apart.

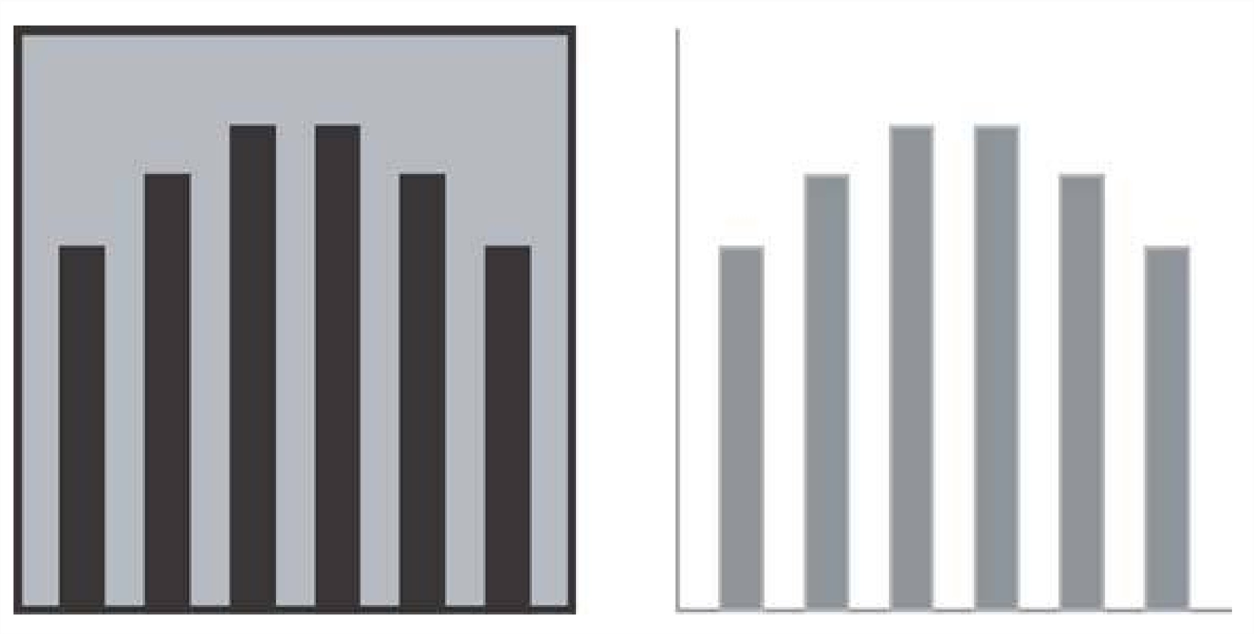


Law of Continuity

points connected by straight or curving lines are seen in a way that follows the smoothest path

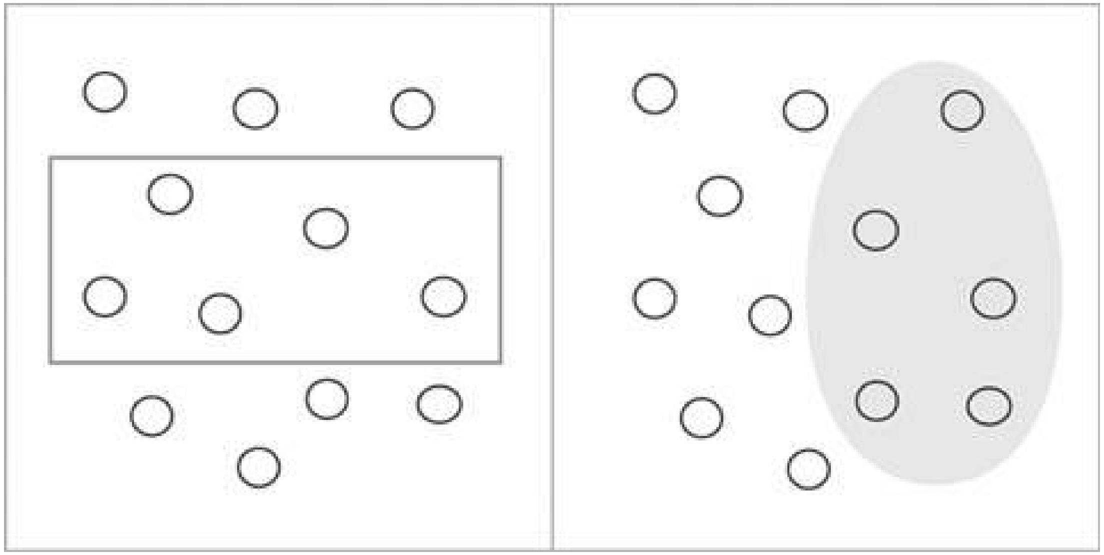
Law of Closure

Elements perceived as belonging to the same group if they seem to complete some entity (Our brains often ignore contradictory information and fill in gaps in information)



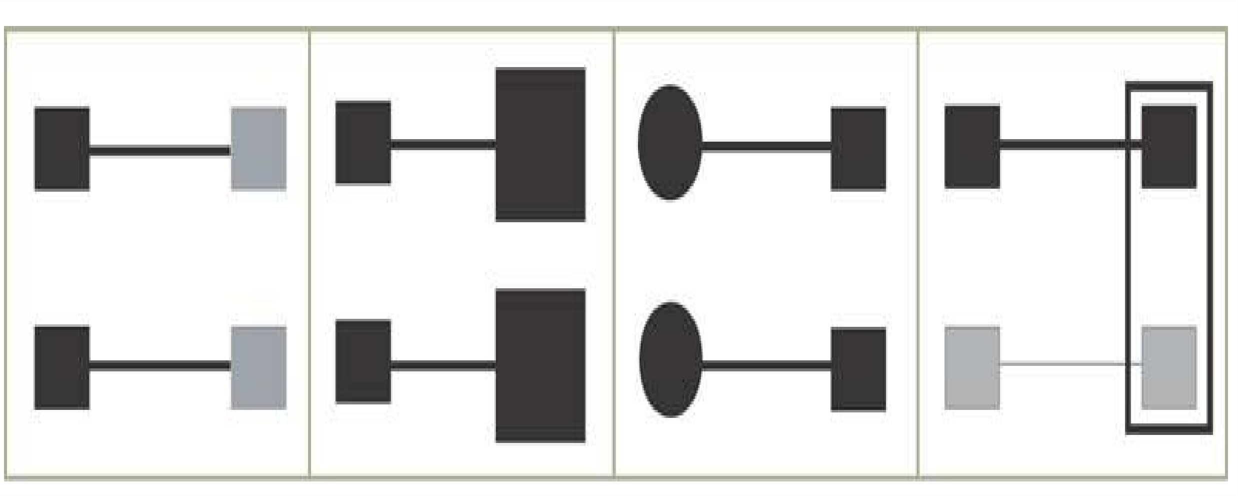
Law of Common Region

Elements located in the same closed region are perceived as belonging to the same group



Law of Connectedness

Elements connected to each other using colors/lines/frames, or other shapes are perceived as a single unit when compared with other elements that are not linked in the same manner.

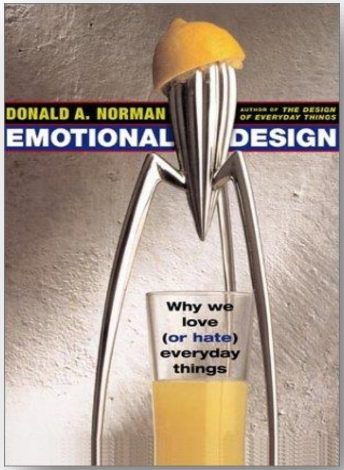


Strong Visual Design

- Use Contrasts, Repetitions, Alignments, Proximity and Other suggested Techniques

- Produce *Esthetic* Visuals - Why ? **Emotional Aspects**

"Attractive things make people feel good, which in turn makes them think more creatively…. making it easier for people to find solutions to the problems they encounter." - Don Norman



Judging Design

Consider Three Aspects:

- **Visceral :**
  - What is the products initial impact or appearance?
- **Behavioral :**
  - How does the object feel to use?
- **Reflective :**
  - What does the object make you think about?
  - What does it say about it’s owner?

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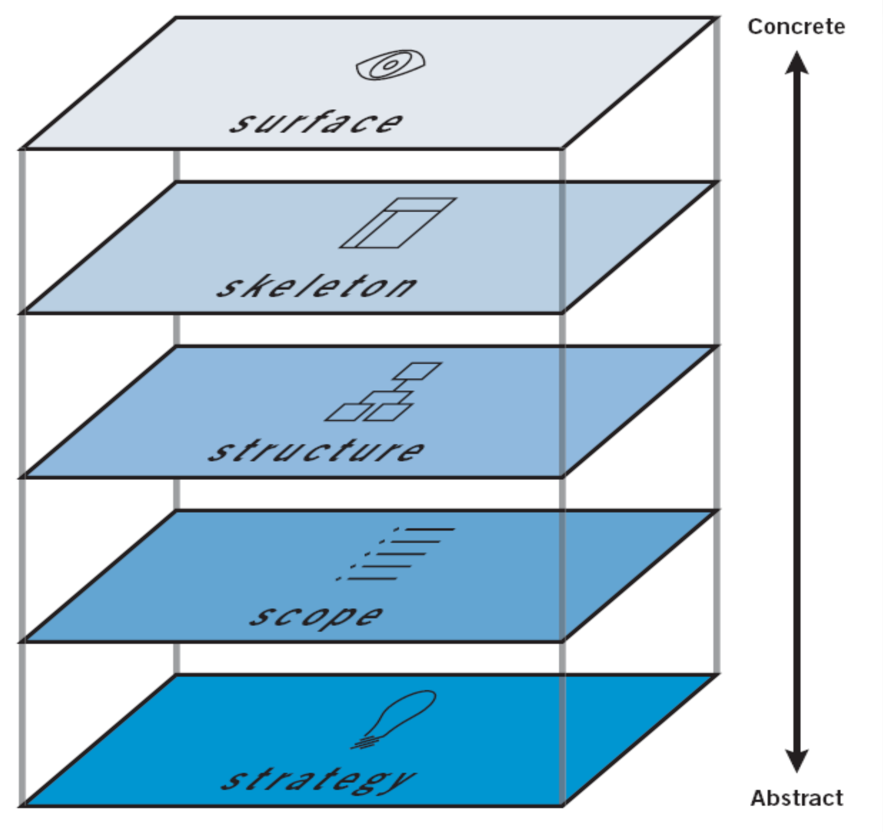
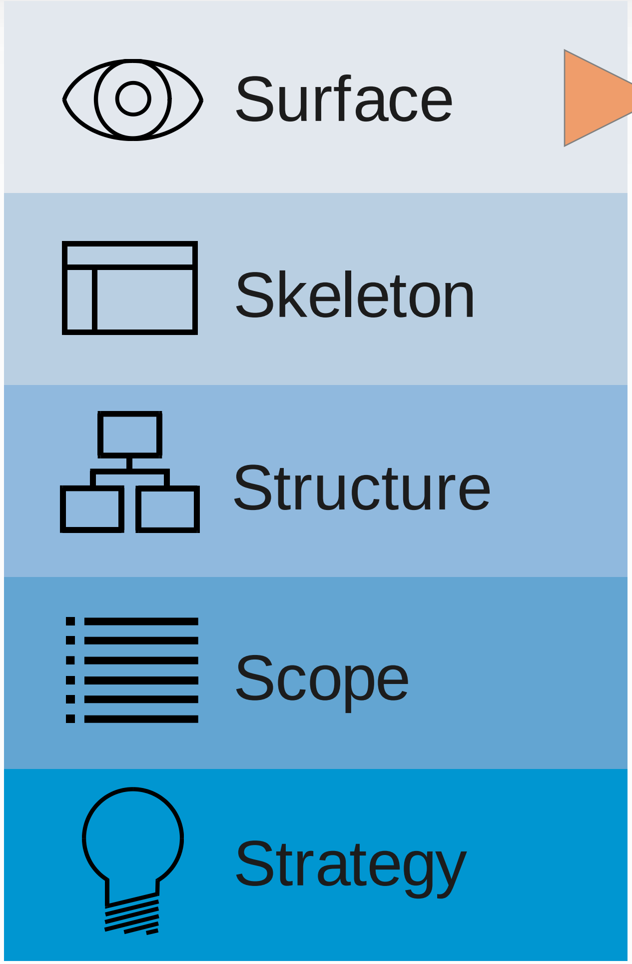
User Experience Layer's Model

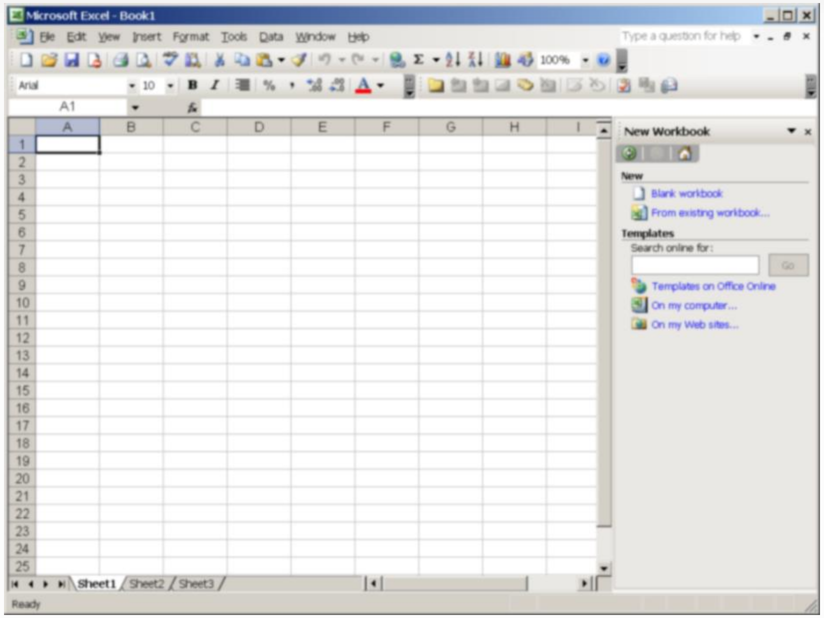
UX 5's Layers



Jesse James Garrett's
Elements of User Experience http://www.jjg.net/elements/

UX Layers - Surface





Surface Layer Describes Finished Visual Design Aspects

pure decoration: visual content that the viewer must process to get to the data

Some colors are hot and demand our attention, while others are cooler and less visible
When colors in two different sections of a dashboard are the same, we are tempted to relate them to one another

10% of males and 1% of females are color-blind (red, yellow, and green)

UX Layers - Skeleton

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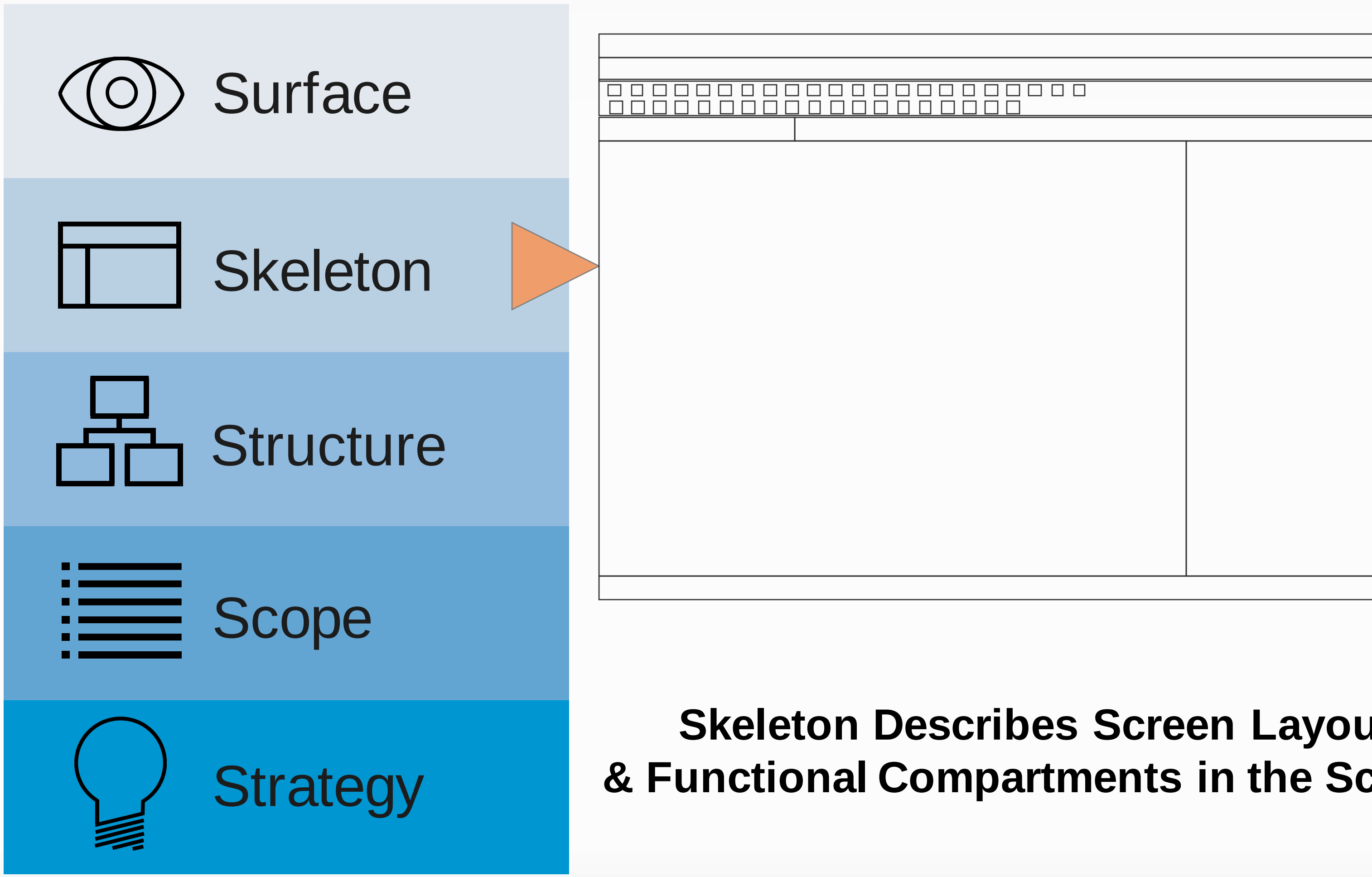
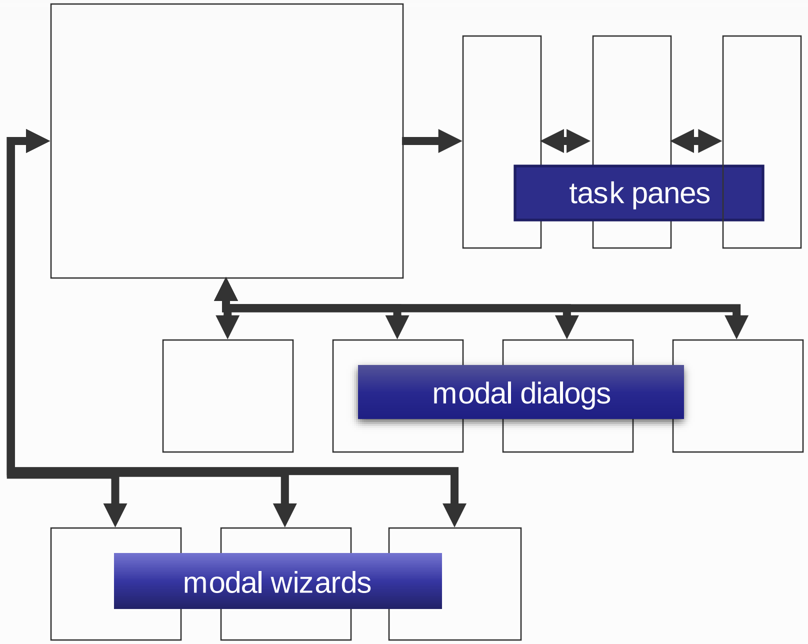


Skeleton Describes Screen Layout
& Functional Compartments in the Screen

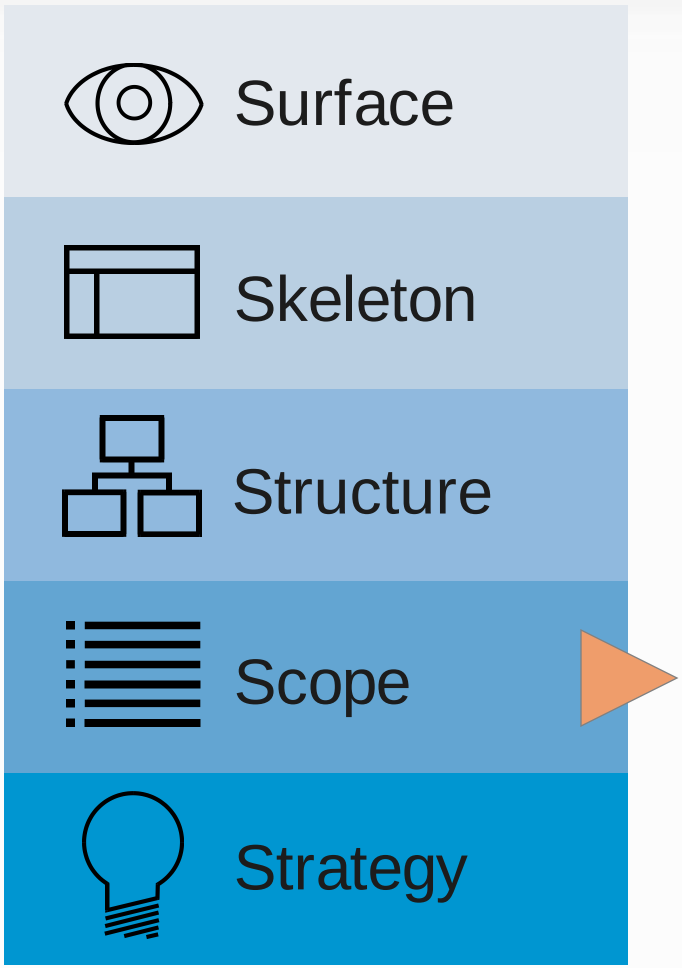
UX Layers - Structure

- Surface
- Skeleton
- Structure
- Scope
- Strategy



Structure Defines Navigation from Place to Place in the User Interface

UX Layers - Scope

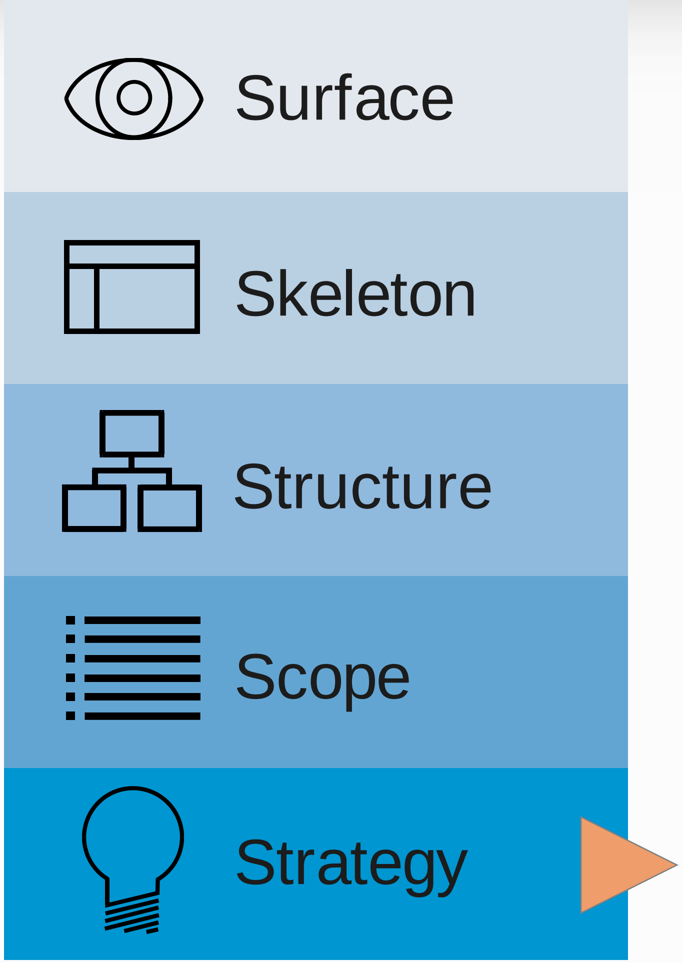


user tasks:

- enter numbers
- enter text
- enter formulas
- format cells
- sort information
- filter information
- aggregate information
- graph data
- save data
- import data
- export data
- print
- …..

The Places in the User Interface Built to Support User Task-Centric Scope

UX Layers - Strategy



business goals:
- displace competitive products
- motivate sale of other integrated products
- establish file format as default information sharing format
- …

user constituencies:
- accountant
- business planner
- housewife
- …

usage contexts:
- office desktop
- laptop on airplane
- Cellular on the way
- …